

■ Coding Flashcard: Cosine Similarity

Step-by-step logic to implement cosine similarity (no libraries):

- 1■ ■ ****Validate inputs**** — both must be lists, same length, and non-empty.
- 2■ ■ ****Compute dot product:**** $\sum(a[i] * b[i])$ for each pair.
- 3■ ■ ****Compute magnitudes:**** $\sqrt{\sum(x^2)}$ for both vectors.
- 4■ ■ ****Check zero magnitude:**** if mag_a or $\text{mag_b} == 0 \rightarrow$ return 0.0.
- 5■ ■ ****Return cosine similarity:**** $\text{dot} / (\text{mag_a} * \text{mag_b})$.

■ **Pro tip:** Round output for readability, e.g. `round(value, 4)`

```
from math import sqrt

def cosine_similarity(a, b):
    if not isinstance(a, list) or not isinstance(b, list):
        raise TypeError("Inputs must be lists.")
    if len(a) != len(b):
        raise ValueError("Same length required.")
    if len(a) == 0:
        raise ValueError("Vectors cannot be empty.")

    dot = sum(float(x) * float(y) for x, y in zip(a, b))
    mag_a = sqrt(sum(float(x)**2 for x in a))
    mag_b = sqrt(sum(float(y)**2 for y in b))
    if mag_a == 0.0 or mag_b == 0.0:
        return 0.0
    return dot / (mag_a * mag_b)
```